

DISCOVERERS

- The Greek philosopher **Democritus**, in the 5th century BCE, set out the idea that everything is made from tiny units called **atoms**.
- In the early 1800s, the British chemist **John Dalton** offered proof that **atoms** existed by showing that gases always spread out evenly.
- In 1895, the French physicist **Antoine Henri Becquerel** discovered that certain materials gave out invisible rays. This was named as **radioactivity** in the early 1900s by the French physicists Marie and Pierre Curie.

- **Sir J J Thomson**, a British physicist, discovered the first subatomic particles in 1897. They are now called **electrons**.
- In 1911, the New Zealand physicist **Baron Ernest Rutherford** led a team that discovered the positively charged core of an atom, which he named the **nucleus**. He later showed that the positive charge was due to a subatomic particle called the **proton**.
- In 1913, the Danish physicist **Niels Bohr** figured out how electrons and protons were **organized in an atom**. This helped scientists to understand how atoms could form **bonds**.

IN POPULAR CULTURE

- In the film *Avatar*, **human miners** are destroying the alien moon Pandora to extract a fictional element called **unobtainium** that is crucial to Earth's survival.
- In Marvel comics and films, **vibranium** is a very strong metal found only in the fictional African country of Wakanda. This metal is used to make **Captain America's** almost indestructible **shield**.
- **Adamantium** is a tough fictional alloy used in the **claws of Wolverine**, one of the X-Men in Marvel comics and films.

- In Middle-earth, the fictional world created by the British author J R R Tolkien, **mithril** is a metal that **shines like silver** and is stronger than steel. Once mined by dwarves, this metal is **priceless** because it is no longer produced. The most famous item made of mithril in Tolkien's works is the mail armour worn by Frodo Baggins, a hobbit.
- In *Star Trek*, a crystalline element called **dilithium** is the central component of a Federation spaceship's warp drive that allows it to travel many times **faster than the speed of light**.



The element lead blocks the X-ray vision of Superman, the famous superhero of DC comics and films.